Equality and universal comparison of positive exponential terms

21 October 2026
Working manuscript

Abstract

We study expressions generated from positive integer variables, the constant 1, addition, and $\mathsf{E}(f,g) = 2^f g$. Universal strict and weak comparison are Π_1^0-complete already at exponential depth two. In contrast, extensional equality is decidable with any finite number of independent variables, and universal comparison is decidable with one independent variable at arbitrary finite depth. The proofs give an existential definition of multiplication, recursive normal forms, and computable eventual-sign bounds. We also transfer the obstruction to bounded two-component vectors in Wilken–Weiermann's assignment syntax. The results refute unrestricted effectiveness of the source's semantic comparison, while leaving open effective restricted comparisons adequate for actual derivation assignments.

1 A comparison problem with two distinct boundaries

An identity asserts that two expressions agree at every input. A universal inequality asserts that their difference never has a counterexample. For the expressions considered here these two tasks have different computability behavior. The distinction persists although every individual evaluation is a finite calculation on positive integers.

Let $\mathcal{F}_d$ be the finite terms generated by

$$1, \quad x_1, \ldots, x_d, \quad f + g, \quad \mathsf{E}(f,g) = 2^f g.$$

Variables range independently over $\mathbb{N}_{>0}$. A positive integer constant is a sum of copies of 1; zero is not a source term. Exponential depth counts nesting of E, with addition taking the maximum of its children's depths. Write $f \prec g$ and $f \preceq g$ for strict and weak inequalities holding at every positive input tuple.

For example, $\mathsf{E}(x+1,y) + \mathsf{E}(x,y)$ and three copies of $\mathsf{E}(x,y)$ have the same value $3y2^x$. Our normal form recognizes this identity without testing inputs. On the other hand, a sum of squares of differences can vanish exactly at the solutions of an arbitrary Diophantine equation. We encode that test using the allowed syntax, with no subtraction, zero, or ordinary variable multiplication in the constructed terms.

There are three main conclusions:

1. universal comparison with an unrestricted finite collection of independent inputs is undecidable even at depth two;

2. equality is decidable uniformly in that finite collection and at any finite depth;

3. with one independent input, strict and weak universal comparison are decidable at any finite depth, and a false comparison has a computable least counterexample.

The number of variables in the undecidability reduction grows with the input. We do not establish a smallest fixed number of variables at which comparison becomes undecidable.

1.1 The source question

Wilken and Weiermann define semantic comparison by quantifying over admissible substitutions and conjecture its effectiveness immediately after their Definition 2.8 [1]. Their definitions allow the fragment above: ground-component substitutions take positive finite values, and every positive integer has a closed representation as a sum of 1's. Natural sum and natural product are ordinary addition and multiplication on these values. Their ordinal norm $\mathrm{N}(n)$ equals n for finite n. Thus their comparison on these terms is exactly comparison over $\mathbb{N}_{>0}^{d}$, not an approximation to it.

The variable multiplier in $\mathsf{E}(f, g)$ matters. We are not studying only addition and the unary operation $f \mapsto 2^f$. The use of *unary* here counts independent numerical inputs.

The undecidability proof uses MRDP as its published input. The identity and one-variable procedures use the elementary separation argument below, rather than a transcendence assumption. The final vector result uses the source's own norm and collapsing-function estimates.

2 Universal comparison at depth two

Theorem 2.1 (Finite depth-two obstruction)**.** *For either $\triangleleft \in \{<, \leq\}$, consider pairs (A, B) in the finite fragment $\mathcal{F}_d$, with d allowed to vary, satisfying*

$$\forall \chi \text{ admissible in Definition 2.7 of [1]}, \qquad \mathrm{val}(A\chi) \triangleleft \mathrm{val}(B\chi)$$

is Π_1^0-complete already when the terms use only 1, independently named ground-type variables, $+$ and E, and have E-nesting depth at most two. Every value is a positive integer, and each output term belongs to every $\mathcal{C}_i$, $i \geq 1$. The same restricted completeness holds for comparison of their norms.

For these variables, the original admissible domain is exactly the set of positive integer tuples. Indeed closed $\mathcal{B}_0$-terms are positive finite, every positive integer is represented by a sum of 1's, and the bounded-norm condition on a one-component substitution is automatic [1, Definitions 2.5–2.7]. All variables introduced below are fresh ground-type variables of this kind.

2.1 A bounded multiplication identity

Lemma 2.2. *Let a, y, s be positive integers and suppose $y < 2^a - 1$ and $s < a(2^a - 1)$. Then $s = ay$ if and only if there exist positive integers u, q such that*

$$2^a u + 1 = u + 2^s, \tag{1}$$

$$2^a q + y + 1 = u + q + 2^a. \tag{2}$$

Proof. Write $B = 2^a$ and $D = B - 1$. Equation (1) says $Du = 2^s - 1$. Division of s by a, say $s = ka + r$ with $0 \le r < a$, gives $2^s - 1 \equiv 2^r - 1 \pmod{D}$. Since $0 \le 2^r - 1 < D$, divisibility forces $r = 0$. Thus $k = s/a \ge 1$ and

$$u = 1 + B + \cdots + B^{k-1} \equiv k \pmod{D}.$$

Equation (2) is $D(q-1) = u - y$, so $k \equiv y \pmod{D}$. The two bounds give $1 \le k, y < D$, whence $k = y$ and $s = ay$.

Conversely, if $s = ay$, take

$$u = \sum_{j=0}^{y-1} B^j, \qquad q = 1 + \frac{u - y}{D}.$$

The second expression is an integer because $u \equiv y \pmod{D}$, and it is positive because $u \ge y$. Both equations follow. This includes $y = 1$, when $u = q = 1$. □

The lemma's bounds and equations are expressible using $1, +, \mathsf{E}$: they contain no product of two arbitrary variables. The apparent products $2^a u$, $2^a q$ and $a2^a$ are exactly permitted exponential-constructor terms.

Lemma 2.3 (Seven-equation definition of multiplication)**.** *For positive x, y, w, put $a = x + y + 1$ and $b = a + x$, treating these as affine term abbreviations. Then $w = xy$ if and only if the following seven source equations have a solution in positive integers s, t, u, v, q, r, h, k:*

$$\begin{aligned} \mathsf{E}(a,u) + 1 &= u + \mathsf{E}(s,1), & \mathsf{E}(b,v) + 1 &= v + \mathsf{E}(t,1), \\ \mathsf{E}(a,q) + y + 1 &= u + q + \mathsf{E}(a,1), & \mathsf{E}(b,r) + y + 1 &= v + r + \mathsf{E}(b,1), \\ s + a + h &= \mathsf{E}(a,a), & t + b + k &= \mathsf{E}(b,b), \\ t &= s + w. \end{aligned}$$

Every side has E-nesting depth at most one.

Proof. Since $a \ge y + 2$ and $b > a$, the bounds $y < 2^a - 1$ and $y < 2^b - 1$ hold identically. The positive slacks h, k give $s < a(2^a - 1)$ and $t < b(2^b - 1)$. Lemma 2.2 therefore forces $s = ay$ and $t = by$. The last equation now gives $w = (b - a)y = xy$.

Conversely, if $w = xy$, choose $s = ay$ and $t = by$; use the witnesses from the preceding lemma for the two pairs of equations, and set

$$h = a(2^a - 1 - y) > 0, \qquad k = b(2^b - 1 - y) > 0.$$

All seven equations hold. Products in these witness formulas are ordinary mathematical calculations; the displayed source equations contain only $1, +, \mathsf{E}$. No disjunction, zero-valued variable or extra quantifier alternation is needed. □

2.2 From Diophantine equations to one universal comparison

Proof of Theorem 2.1. Solvability of integer-coefficient polynomial equations over positive integers is Σ_1^0-complete by the MRDP theorem [2]. The positive convention follows from the usual nonnegative convention by replacing each nonnegative variable with $x - 1$ and expanding the polynomial.

Given such a polynomial P, separate its positive and negative coefficient parts and express

$$P_+(\mathbf{x}) + 1 = P_-(\mathbf{x}) + 1.$$

Each side has a finite arithmetic circuit using positive constants, addition and multiplication. Delete zero monomials; the added 1 ensures that both sides are positive, even if one part was empty. Introduce a fresh positive variable for every circuit wire. Addition and constant gates give affine equations. Replace each multiplication gate by Lemma 2.3, using fresh auxiliaries. Retain the final output equality. We obtain a finite system

$$f_i(\mathbf{z}) = g_i(\mathbf{z}), \qquad 1 \le i \le m,$$

over positive variables which has a solution exactly when P does. All its terms use only $1, +, \mathsf{E}$, and every f_i, g_i has E-depth at most one. This is an effective syntactic construction; the new variables, like the original ones, become independently substitutable ground-type variables.

Form the source terms

$$U = 1 + \sum_{i=1}^{m} \big(\mathsf{E}(f_i + f_i, 1) + \mathsf{E}(g_i + g_i, 1)\big),$$
$$V = 1 + \sum_{i=1}^{m} \mathsf{E}(f_i + g_i + 1, 1).$$

At every positive tuple, ordinary integer arithmetic gives

$$U - V = \sum_{i=1}^{m} \left(2^{f_i} - 2^{g_i}\right)^2 \ge 0. \tag{3}$$

The right side is a proof identity, not source syntax. It vanishes exactly when all $f_i = g_i$. Consequently

$$P \text{ has no positive zero} \quad \Longleftrightarrow \quad V \prec U \quad \Longleftrightarrow \quad V + 1 \preceq U.$$

Here the final equivalence uses that the nonnegative difference is an integer. These are computable many-one reductions from a Π^0_1-complete problem, with output depth at most two.

For the upper bound on this restricted family, enumerate all positive integer substitutions and evaluate the finite terms exactly. A failed comparison is witnessed by one enumerated tuple, so the relation is co-c.e. This establishes the claimed upper bound on the finite fragment.

For every $i \ge 1$, simultaneous induction on these finite source terms gives membership in $\mathcal{C}_i$: ground variables lift from $\mathcal{C}_0$, sums preserve membership, and $\mathsf{E}(f, g)$ uses $f \in \mathcal{C}_{i+1}$ and $g \in \mathcal{C}_i$ [1, Definition 2.9]. Finally all values are finite, so $\mathrm{N}(U) = U$ and $\mathrm{N}(V) = V$. The identical reduction therefore proves the asserted norm-comparison completeness. □

2.3 The obstruction persists on bounded $\mathcal{C}$-vectors

Theorem 2.4 (Two-component bounded-vector obstruction)**.** *Both strict and weak comparison in Definition 2.8 are undecidable even on a computably constructed family of two-component $\mathcal{C}$-vectors whose bounded norm is proved uniformly. The family gives Π^0_1-hardness while retaining the finite depth-two terms as higher components.*

Proof. Use the U, V just constructed and put $R = U + V + 1$. Let $L \geq 1$ be the number of nodes in the source term tree of R, and let $z_1, \ldots, z_n$ list all its variables. Set

$$s_0 = 1 + \sum_{j=1}^{n} z_j, \qquad b = (L+1)s_0, \qquad g = 5b.$$

These are positive affine source terms; the integer coefficients abbreviate finite sums. We claim $R \preceq F_2(g)$.

Fix a substitution. Every leaf value is at most s_0 and every depth-zero subtree has value at most b. Flatten sums at each depth. There are at most L summands, so every depth-one subtree is bounded by

$$Lb2^b \leq 2^{3b}.$$

At depth two this gives

$$R \leq L\, 2^{2^{3b}+3b} \leq 2^{2^{3b}+4b} \leq 2^{2^{5b}}.$$

We used $L, b \leq 2^b$ and $2^{3b} + 4b \leq 2^{5b}$ for $b \geq 1$. The source defines $F_0(n) = 2^n$ and $F_{j+1}(n) = F_j^{n+1}(n)$, so $F_2(g) \geq F_1(g) \geq 2^{2^g}$ [1, p. 15]. This proves the claim over exactly the admissible substitution domain.

Now define the source term

$$H = \psi(\omega \cdot R + g).$$

We have $R \in \mathcal{C}_1$, $g \in \mathcal{C}_0$ and $\mathrm{N}(R) = R \preceq F_2(g)$, which is precisely the side condition for $H \in \mathcal{C}_0$. Under each substitution, natural product gives $\omega \otimes R = \omega \cdot R$, and hence

$$H \geq \mathrm{N}(\omega \otimes R \oplus g) = 2R + g \geq R$$

by [1, Proposition 2.3]. Therefore $(H, V+1)$, (H, U) and $(H+1, U)$ are bounded $\mathcal{C}$-vectors. Definition 2.8 gives exactly

$$(H, V+1) \preceq (H, U) \quad \Longleftrightarrow \quad V + 1 \preceq U \quad \Longleftrightarrow \quad (H, V+1) \prec (H+1, U),$$

because $H \prec H + 1$ holds under every substitution. Applying (3) proves the reduction. □

This result directly concerns the original scalar and vector comparison relations, including bounded $\mathcal{C}$ inputs. It does not establish that the hard vectors occur as assignments of actual T-terms, and it does not rule out effective sufficient comparisons or a differently designed termination ordering. It rules out an algorithm deciding all the stated semantic comparisons exactly.

3 Normal forms and extensional equality

The auxiliary normal-form syntax permits zero. Define, by finite well-founded trees, expressions

$$F = \sum_{e \in \mathrm{supp}(F)} P_e(\mathbf{x}) 2^e, \qquad P_e = c_e + \sum_{i=1}^{d} a_{e,i} x_i, \tag{N1}$$

where every P_e is a nonzero affine polynomial with nonnegative integer coefficients, the support is finite, and each key e is itself a normal form of smaller height. Every key is *constant-free*: its coefficient of 1 in the zero-exponent term is zero. Zero is represented by the empty

sum. Equal keys are merged, and a zero coefficient is deleted. A normal form is a finite map, so the display order of its keys has no mathematical significance.

For any normal form F, let $c(F)$ denote the constant coefficient of its zero-exponent term, with value zero when that term is absent. Let F° be the form obtained by removing just that coefficient. Thus $F = c(F) + F^\circ$ and F° is constant-free. The name "constant-free" is syntactic: for example, 2^{x_1} has $c(F) = 0$ although its value at $x_1 = 0$ would be 1.

Lemma 3.1 (Effective normalization)**.** *Every term in $\mathcal{F}_d$ has an effectively computable normal form* $\mathrm{nf}(f)$ *of the shape* (N1)*, with the same value at every positive integer tuple. All operations terminate by finite syntax recursion.*

Proof. The normal forms of 1 and x_i have only the zero key. Addition merges the two finite maps and adds their affine coefficients. Suppose F, G are the normal forms of f, g, and write $F = c + R$, with R constant-free. Then use

$$2^F G = \sum_{e \in \mathrm{supp}(G)} (2^c P_e)\, 2^{R+e}. \tag{N2}$$

Addition of the two already normalized constant-free forms R, e gives a constant-free key. Merge equal resulting keys. The coefficients remain affine because only addition and multiplication by a numerical constant occurred. The identities used are ordinary integer distributivity and $2^{u+v} = 2^u 2^v$. Keys remain of smaller finite height than the resulting outer sum. Computing 2^c may be expensive, but it is a finite computation; no complexity bound is being asserted. □

3.1 A separation lemma without transcendence assumptions

Lemma 3.2 (Finite-family ray separation)**.** *Given a finite collection $\mathcal{S}$ of normal forms, close it under taking keys and add zero. One can effectively choose positive integers $a_1, \ldots, a_d$ such that, on the ray $x_i = a_i t$ with integer $t > 0$, the following statements hold for all $F, G \in \mathcal{S}$.*

1. *If $F \neq G$, their difference is eventually nonzero with a fixed sign.*
2. *If F, G are constant-free and distinct, then $|F(\mathbf{a}t) - G(\mathbf{a}t)| \geq t$ for all sufficiently large t.*

The common thresholds and the eventual signs are computable.

Proof. Include the zero affine coefficient when collecting the affine coefficient vectors of all forms in $\mathcal{S}$. Choose a positive integer B so large that, for every difference v of two collected vectors of variable coefficients,

$$B > 1 + \sum_i |v_i|.$$

Set $a_i = B^{i-1}$. If $v \neq 0$ and j is its highest nonzero index, the absolute value of $v_j a_j$ exceeds the sum of the absolute values of the earlier contributions. Thus $v \cdot \mathbf{a} \neq 0$. Consequently every nonzero difference of two collected affine coefficients becomes a nonzero affine polynomial in t. A difference with nonzero variable part acquires a nonzero integer slope.

Induct on normal-form height, simultaneously for pairs in the finite collection. When only the zero key occurs, there is an affine polynomial $At + C$. If $A \neq 0$, its eventual sign is the sign of A, and $t \geq |C| + 1$ suffices to make it nonzero. If $A = 0$, its nonzero constant sign is

immediate. When both forms are constant-free, $C = 0$, so a nonzero difference has absolute value at least t.

For the induction step, subtract the two maps and discard zero coefficients. Distinct remaining keys are constant-free forms of smaller height. By induction their pairwise eventual order is strict, and any ordered pair has a gap at least t after a computable threshold. Let e_* be the greatest remaining key in that eventual order, with coefficient $P_*(t) = A_*t + C_*$. For all sufficiently large t, $P_*(t)$ has a fixed sign and integer absolute value at least one. Put

$$C = \sum_{e \neq e_*} (|A_e| + |C_e|).$$

Once every key gap $e_*(t) - e(t)$ is at least t, the sum of the absolute values of the lower contributions is at most

$$Ct\, 2^{e_*(t)-t}. \tag{N3}$$

For $t \geq \max(4, 2C)$, the elementary inequality $2^t \geq t^2$ implies $Ct2^{-t} \leq 1/2$. Thus the leading summand determines the sign of the difference.

If $e_* = 0$, there were no other surviving keys, since every nonzero constant-free key is eventually positive. This is the already treated affine case. If $e_* > 0$, induction applied to e_* and zero gives $e_*(t) \geq t$ after another computable threshold. The difference then has absolute value at least $2^{e_*(t)-1} \geq t$. This proves both assertions. Taking maxima of the finitely many recursive thresholds makes every step effective. No algebraic-independence or transcendence theorem is used. □

Theorem 3.3 (Decidable extensional equality)**.** *For any finite number of independent positive ground variables and any finite exponential nesting depths,*

$$(\forall \mathbf{n} > 0)\ f(\mathbf{n}) = g(\mathbf{n}) \quad \Longleftrightarrow \quad \mathrm{nf}(f) = \mathrm{nf}(g). \tag{N4}$$

Consequently extensional equality in $\mathcal{F}_d$ is decidable, uniformly in d and the two input terms. If the normal forms differ, a separating positive integer tuple can be computed.

Proof. Equal normal forms imply equality by normalization soundness. For unequal forms, apply Lemma 3.2 to the two forms and their keys. Their values differ on every sufficiently large point of the computed positive ray. The constructive proof supplies such a point. Structural equality of the finite normal forms is decidable. □

3.2 A complete universal comparison procedure with one variable

Theorem 3.4 (One-variable universal comparison)**.** *Let $f, g \in \mathcal{F}_1$. Both*

$$(\forall n > 0)\ f(n) < g(n), \qquad (\forall n > 0)\ f(n) \leq g(n) \tag{N5}$$

are decidable, uniformly at arbitrary finite exponential depth. A false instance has an effectively computable least positive counterexample.

Proof. Normalize f, g. Equality is handled immediately. With one variable the ray $x = t$ already separates every nonzero affine coefficient difference, so the proof of Lemma 3.2 applies with $a_1 = 1$. It computes a sign $s \in \{-1, 1\}$ and integer $N \geq 1$ such that $\mathrm{sign}(f(n) - g(n)) = s$ for all $n \geq N$.

For clarity, one permissible threshold recurrence is explicit. For a nonzero affine difference $An + C$, take $N = 1$ if $A = 0$, and $N = |C| + 1$ otherwise. For a difference whose greatest surviving key e_* is nonzero, take the maximum of:

1. 4 and $2\sum_{e\neq e_*}(|A_e|+|C_e|)$;
2. $|C_*|+1$ if $A_*\neq 0$, and 1 otherwise;
3. the recursively obtained thresholds at which $e_*(n)-e(n)\geq n$ for every other surviving key e, and $e_*(n)\geq n$.

Keys are constant-free, so their distinct-pair recursive differences always have the gap property required in the third item. The estimates in the ray-separation proof verify this recurrence. It is a recursion on strictly smaller key heights, not a search for an unspecified asymptotic bound.

Evaluate the two terms exactly at $1,\ldots,N-1$. If one of these is a counterexample, return the first. If none is, the eventual sign either proves the comparison for all remaining integers or makes N its least counterexample. Every integer evaluation terminates. This proves a total computability theorem, without claiming a practical bound on the magnitudes of the evaluated integers. □

Corollary 3.5 (Further exact consumers)**.** *The same procedures decide equality of finite-valued ordinal vectors over any finite set of independent ground variables, and strict/weak vector comparison and bounded norm for finite-valued $\mathcal{O}$-vectors with one independent ground variable. They also decide comparison after any fixed affine one-parameter substitution with nonnegative integer coefficients and positive coordinate values for finitely many ground variables: explicitly, $x_i = a_i n + b_i$ with $a_i, b_i \in \mathbb{N}$ and $a_i + b_i > 0$.*

Proof. Vector equality and comparisons are the componentwise combinations specified by Wilken–Weiermann Definition 2.8. On finite values the norm equals the value. Such an affine substitution is a term built from the remaining variable, positive integer constants and addition; substitution therefore reduces its comparison to Theorem 3.4. A zero constant summand is omitted. A coordinate that is a positive constant is allowed as well. □

4 What the algorithms do and do not decide

The normal form is an identity test, not a universal multivariable ordering: a ray witnessing a nonidentity does not show that an inequality holds away from that ray. This is why the identity algorithm is consistent with the depth-two lower bound. With one variable there are only a finite initial segment and the eventual tail, so the same asymptotic argument becomes a complete comparison procedure.

A fixed substitution $x_i = a_i n + b_i$ with nonnegative integer coefficients and $a_i + b_i > 0$ also reduces comparison to the unary case. Zero coefficients are handled by omitting summands; a positive constant coordinate is permitted. This is a restriction to a prescribed path through the input domain, not a decision for independent variables.

The explicit threshold calculations can produce enormous integers. They prove total computability, not a useful running-time estimate. The bounded-vector reduction likewise proves hardness for the specified source syntax, without asserting that the hard vectors are realized by assignments of actual terms of Gödel's T. An algorithm for a justified assignment image, or a sound incomplete comparison adequate for termination proofs, is a distinct target. None of the results constructs an exact transitive termination ordering with a matching ordinal height.

Preparation and status. This manuscript was prepared with AI assistance from a research programme directed by the repository owner. The displayed unbounded arguments, rather than numerical trials, support the theorems. MRDP and the specified Wilken–Weiermann statements are cited inputs. No independent human review, proof-assistant verification, or historical priority claim is asserted. The canonical manuscript and build files are included; no private proof notes or project conventions are required.

References

[1] G. Wilken and A. Weiermann, *Derivation Lengths Classification of Gödel's T Extending Howard's Assignment*, Logical Methods in Computer Science 8(1:19) (2012), 1–44. Source version arXiv:1203.0115v2, Definitions 2.4–2.9 and Proposition 2.3. `https://doi.org/10.2168/LMCS-8(1:19)2012`.

[2] Y. V. Matiyasevich, *Hilbert's Tenth Problem*, MIT Press, 1993. The MRDP theorem is used for Diophantine solvability; the reduction from that problem to exponential comparison is proved here.